

Experiment 14

Demonstrate how to work collaboratively in Git/GitHub on a project using the fork-and-pull request workflow.

Tasks:

1. Fork an existing public GitHub repository (e.g., a sample JavaScript or Python project).
2. Clone the forked repository to your local machine using Git.
3. Create a new branch for the feature or change you want to work on.
4. Make modifications or add new features (e.g., add a function, fix a bug, or update the README).
5. Commit your changes and push the branch to GitHub.
6. Go to the GitHub repository and create a pull request to merge your feature branch into the main branch.
7. Review the pull request and provide feedback on the changes.
8. Respond to feedback by making additional commits to the feature branch if necessary.
9. Once the pull request is approved, merge it into the main branch.
10. Finally submit the link to the pull request along with a summary of the changes you made and how you collaborated.

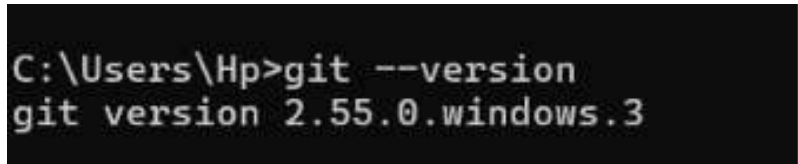
Step 1: Install Git

Git is a version control software that allows us to track changes in our project.

1. Download Git from the official website.
2. Install it using the default settings.
3. Open **Command Prompt**.
4. Check whether Git is installed by typing:

`git --version`

If Git is installed successfully, it displays the Git version.



```
C:\Users\Hp>git --version
git version 2.55.0.windows.3
```

Step 2: Configure Git

Git needs your name and email address so that every commit is associated with your identity.

Run the following commands:

```
git config --global user.name "AKTCHAYA A"
```

```
git config --global user.email "your_email@gmail.com"
```

To verify the configuration:

`git config --list`

If your name and email appear, the configuration is successful.

```
C:\Users\Hp>git config --global user.name "AKTCHAYA A"

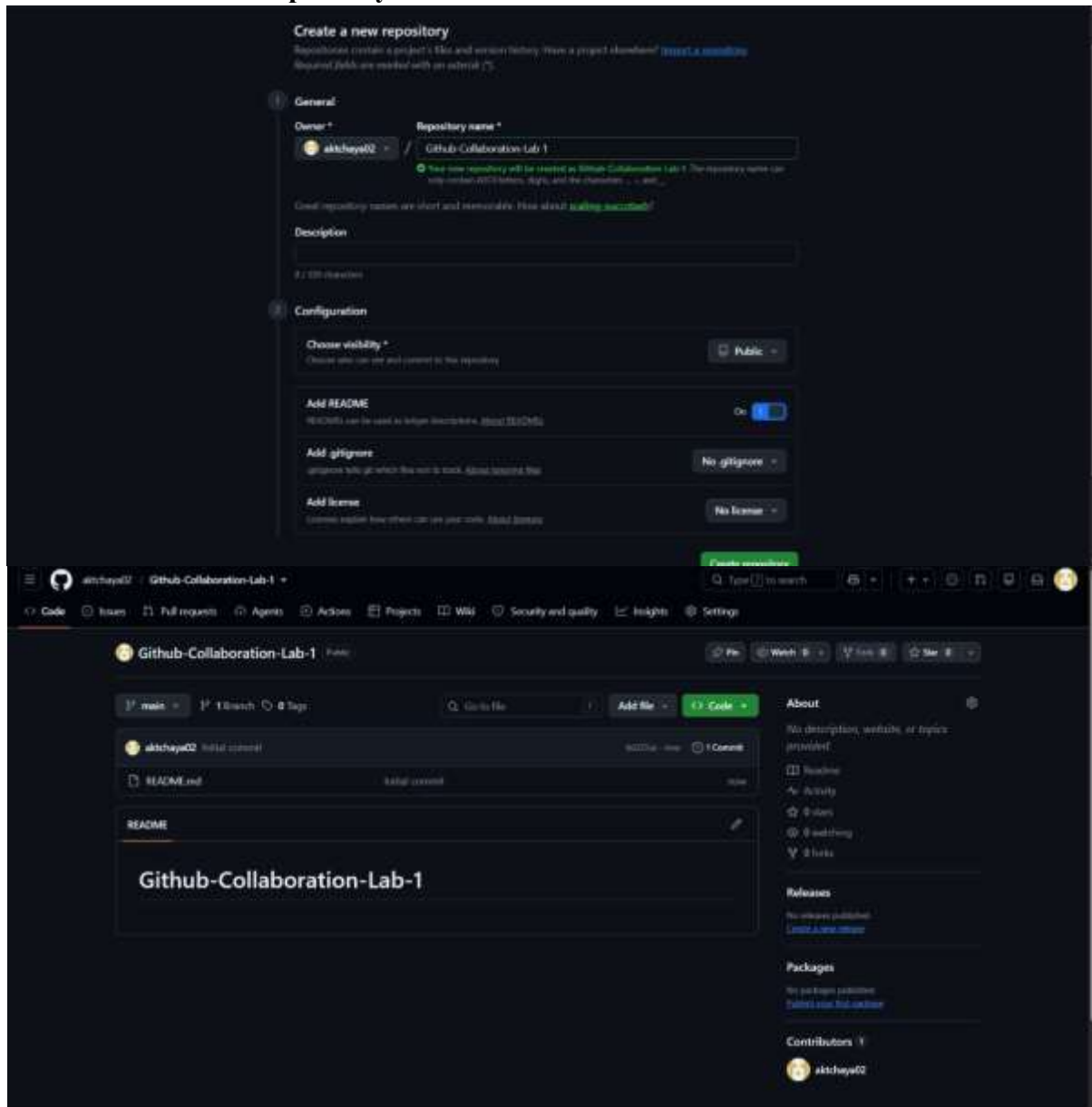
C:\Users\Hp>git config --global user.email "aktchayaathimulam0@gmail.com"

C:\Users\Hp>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=AKTCHAYA A
user.email=aktchayaathimulam0@gmail.com
```

Step 3: Create a New GitHub Repository

1. Log in to GitHub.
2. Click the + icon.
3. Select **New Repository**.
4. Enter the repository name:
Github-Collaboration-Lab
5. Select **Public**.
6. Tick **Add a README.md file**.

7. Click Create Repository



Step 4: Clone the Repository

Cloning means downloading a copy of the GitHub repository to your computer.

1. Click the **Code** button.
2. Copy the HTTPS URL.
3. Open **Command Prompt**.
4. Execute:

```
git clone https://github.com/YourUsername/Github-Collaboration-Lab.git
```

5. Move into the repository folder:

```
cd Github-Collaboration-Lab
```

6. Display the files:

`dir`

The README.md file will be displayed

```
C:\Users\Hp>git clone https://github.com/aktchaya02/Github-Collaboration-Lab-1.git
Cloning into 'Github-Collaboration-Lab-1'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Hp>cd Github-Collaboration-Lab-1

C:\Users\Hp\Github-Collaboration-Lab-1>dir
Volume in drive C is Windows
Volume Serial Number is 6E6C-91A1

Directory of C:\Users\Hp\Github-Collaboration-Lab-1

07-08-2026  13:44    <DIR>          .
07-08-2026  13:44    <DIR>          ..
07-08-2026  13:44                28 README.md
               1 File(s)                28 bytes
               2 Dir(s)  22,181,134,336 bytes free
```

Step 5: Create a New Branch

Instead of working directly on the main branch, create a new branch to develop your changes safely.

Execute:

`git checkout -b feature-update`

Check the available branches:

`git branch`

The current branch is marked with an asterisk (*).

```
C:\Users\Hp\Github-Collaboration-Lab-1>git checkout -b feature-update
Switched to a new branch 'feature-update'

C:\Users\Hp\Github-Collaboration-Lab-1>git branch
* feature-update
  main
```

Step 6: Modify the Project

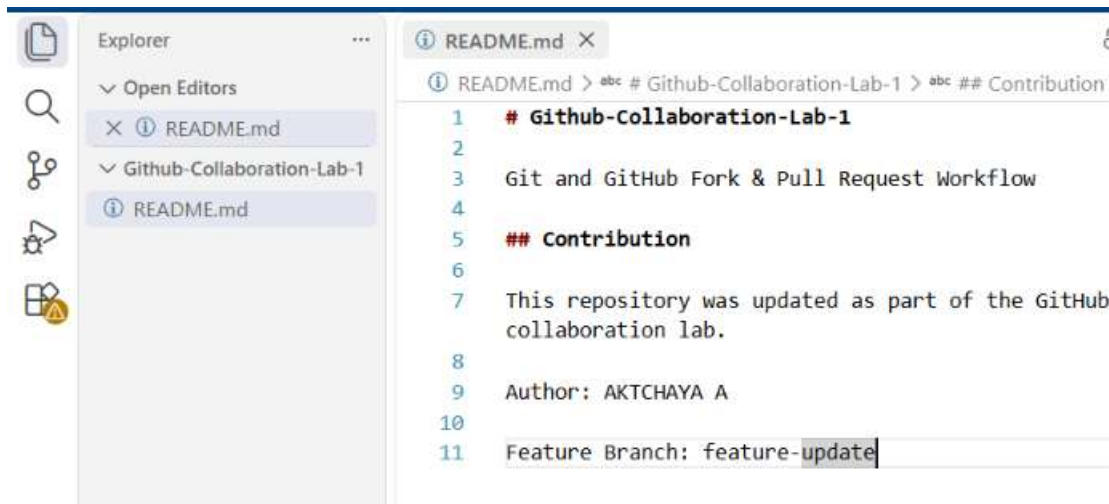
1. Open the repository in **Visual Studio Code**.
2. Open the **README.md** file.
3. Add the following content:

`## Contribution`

This repository was updated as part of the GitHub collaboration lab.

Author: AKTCHAYA A

4. Feature Branch: feature-update
5. Save the file using **Ctrl + S**.



Step 7: Check the Repository Status

Before saving the changes into Git, check the repository status.

Execute:

```
git status
```

Git displays:

```
modified: README.md
```

This means the file has been modified but has not yet been added for commit.

```
C:\Users\Hp\Github-Collaboration-Lab-1>git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

Step 8: Stage the Modified File

Stage the modified file using:

```
git add README.md
```

Check the status again:

```
git status
```

Now Git displays:

```
Changes to be committed
```

This indicates that the file is ready to be committed.

```
C:\Users\Hp\Github-Collaboration-Lab-1>git add README.md

C:\Users\Hp\Github-Collaboration-Lab-1>git status
On branch feature-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

Step 9: Commit the Changes

Save the staged changes permanently into Git.

Execute:

```
git commit -m "Updated README with collaboration details"
```

Git creates a new commit with the specified message.

```
C:\Users\Hp\Github-Collaboration-Lab-1>git commit -m "Update README with col
laboration details"
[feature-update 26a8cdf] Update README with collaboration details
 1 file changed, 11 insertions(+), 1 deletion(-)
```

Step 10: Push the Changes to GitHub

Upload the committed changes to GitHub.

Execute:

```
git push origin feature-update
```

Git uploads the new branch to GitHub.

```
C:\Users\Hp\Github-Collaboration-Lab-1>git push origin feature-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:   https://github.com/aktchaya02/Github-Collaboration-Lab-1/pull/new/feature-update
remote:
To https://github.com/aktchaya02/Github-Collaboration-Lab-1.git
 * [new branch]      feature-update -> feature-update
```

Step 11: Create a Pull Request

1. Open the GitHub repository.
2. Click **Compare & pull request**.
3. Enter:

Title

Updated README with collaboration details

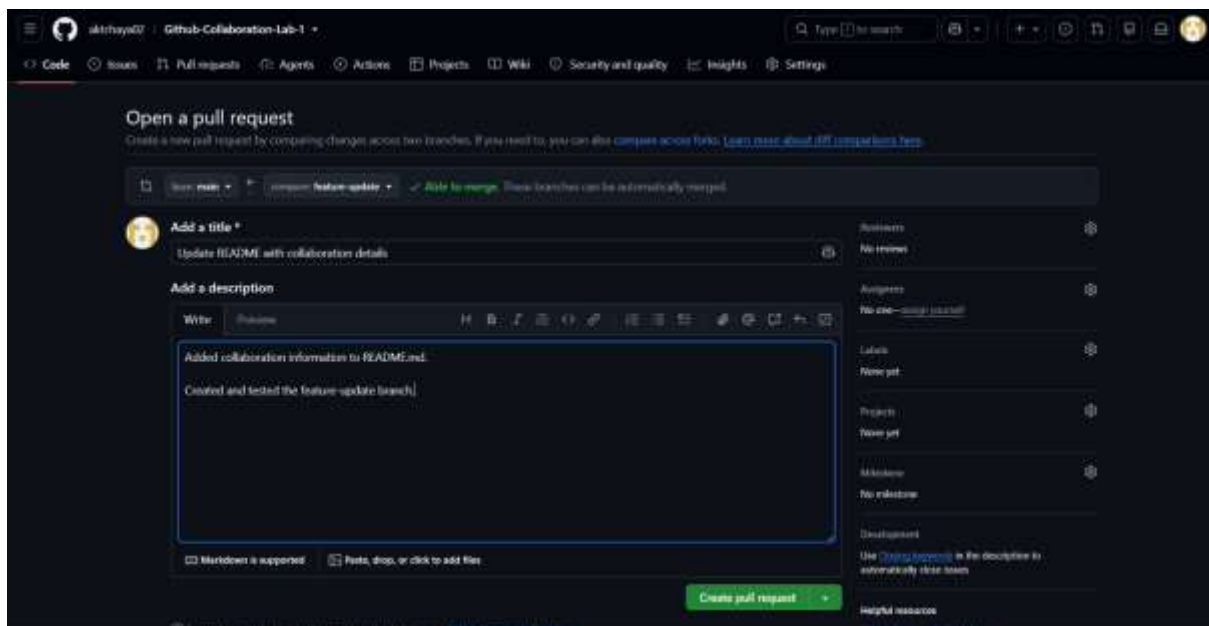
Description

Added collaboration information to README.md.

Created and tested the feature-update branch.

4. Click **Create Pull Request**.

A Pull Request is created.

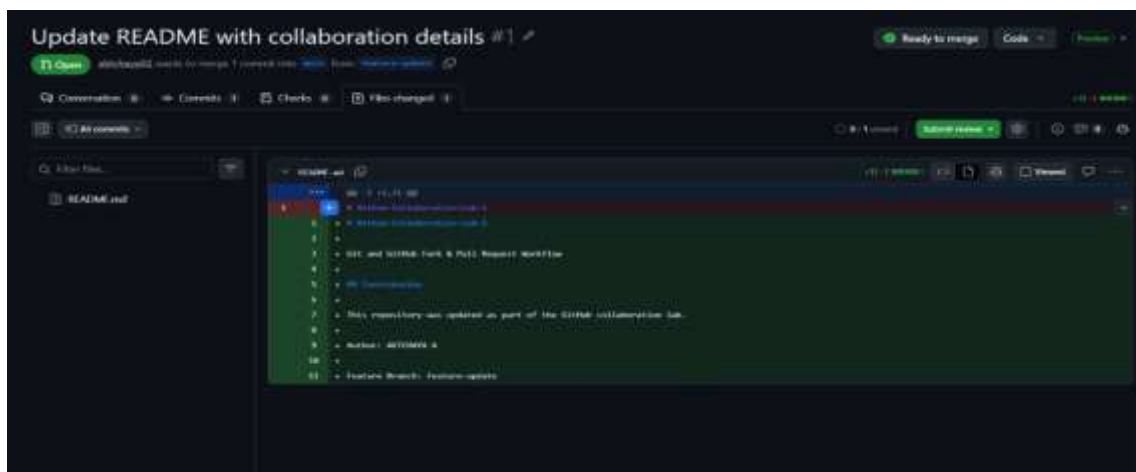


Step 12: Review the Changes

Click the **Files changed** tab.

Review the modifications made to the project.

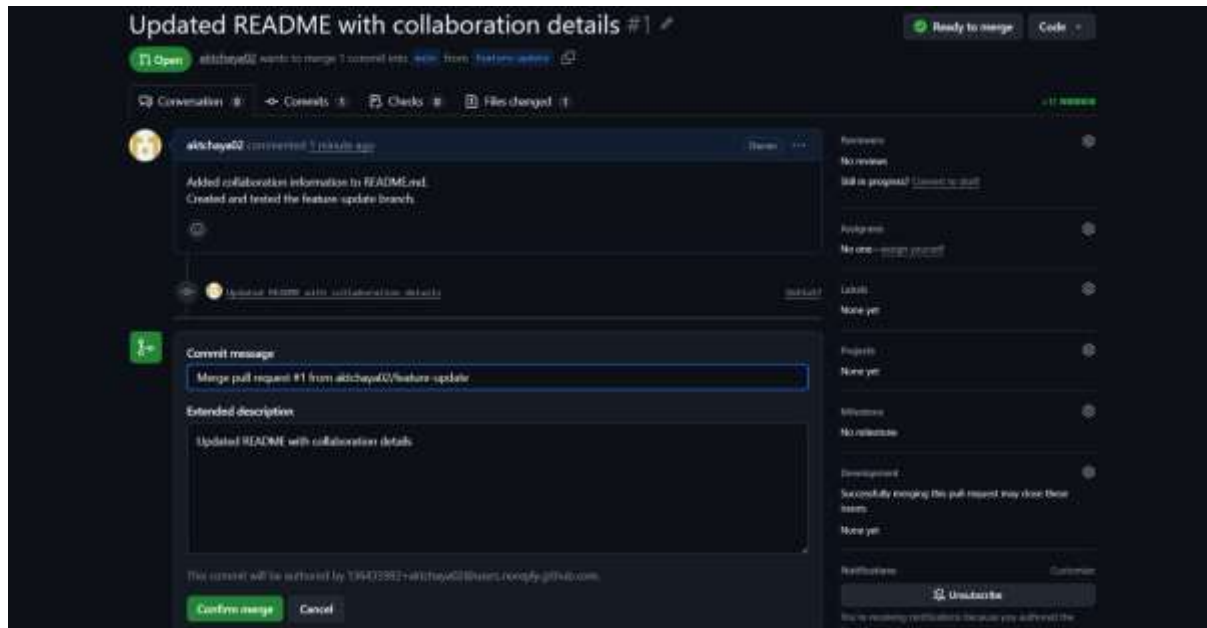
The green highlighted lines represent the newly added content.



Step 13: Merge the Pull Request

1. Open the **Conversation** tab.
2. Scroll to the bottom.
3. Click **Merge Pull Request**.
4. Click **Confirm Merge**.

GitHub merges the feature branch into the main branch.



Step 14: Pull Request Successfully Merged

After merging, GitHub displays the message:

Pull request successfully merged and closed

This confirms that the changes have been merged into the main branch.

You may also delete the feature branch by clicking **Delete Branch**.

